

Workflow

CRM/ERP Platform Documentation

Workflow Automation

Workflows automate actions when records change. They run in the background via BullMQ.

Concepts

Term	Description
Workflow	A named automation rule: trigger + conditions + actions
Trigger	The event that starts the workflow
Conditions	Optional filters — workflow only runs if these pass
Actions	What to do (ordered list)
Execution	A log of one workflow run (stored in <code>WorkflowExecution</code>)

Triggers

Trigger	When it fires
<code>RECORD_CREATED</code>	New record created in the module
<code>RECORD_UPDATED</code>	Any field updated
<code>FIELD_CHANGED</code>	Specific field changes to a specific value
<code>RECORD_DELETED</code>	Record soft-deleted

Trigger	When it fires
<code>SCHEDULED</code>	Time-based (cron expression in <code>triggerConfig</code>)
<code>MANUAL</code>	Triggered explicitly from the UI or API
<code>WEBHOOK</code>	Incoming HTTP webhook call
<code>FORM_SUBMITTED</code>	A linked form receives a submission

Conditions

Conditions are JSON arrays applied before actions run:

```
[
  { "field": "status", "operator": "equals", "value": "approved" },
  { "field": "amount", "operator": "greaterThan", "value": 10000 }
]
```

Operators: `equals` , `notEquals` , `contains` , `startsWith` , `greaterThan` , `lessThan` , `isEmpty` , `isNotEmpty` , `in` , `notIn` .

Action Types

Type	What it does
<code>SET_FIELD</code>	Update a field on the current record
<code>UPDATE_RECORD</code>	Update fields on a related/linked record
<code>SEND_NOTIFICATION</code>	Push in-app notification to users/roles/departments
<code>ASSIGN_USER</code>	Set a user assignment field on the record
<code>TAG</code>	Add/remove tags from the record
<code>CREATE_RECORD</code>	Create a new record in the same or another module

Type	What it does
<code>CHANGE_STATUS</code>	Update a STATUS field to a specific value
<code>SEND_EMAIL</code>	Send email via SMTP using a template or inline body
<code>WEBHOOK</code>	HTTP POST to an external URL with record data
<code>UPDATE_RELATED</code>	Update records in a related module
<code>CREATE_TASK</code>	Create a workspace task
<code>DELAY</code>	Wait N minutes/hours/days before the next action
<code>TRIGGER_WORKFLOW</code>	Start another workflow (chaining)

Each action has a `config` JSON object with action-specific parameters and an `order` integer for sequencing.

SEND_EMAIL Action Config

```
{
  "templateId": "template-id",    // optional: use existing template
  "subject": "Your {{name}} record was updated",
  "body": "<p>Dear {{name}}, your record status is now {{status}}.</p>",
  "recipientType": "field",       // "field" | "users" | "roles" | "email"
  "recipientField": "email",      // field name that holds the recipient email
  "mergeFields": ["name", "status", "id"]
}
```

Merge tags use `{{fieldName}}` syntax and are resolved from the record's data at execution time.

Repeatable Workflows

`isRepeatable: Boolean` controls whether a workflow fires every time its trigger occurs, or only once per record. Set to `false` for one-time actions like "send welcome email on creation."

Workflow Execution Log

Each run creates a `WorkflowExecution` row:

```
{
  "workflowId": "...",
  "recordId": "...",
  "status": "completed",    // pending | running | completed | failed
  "input": { ... },        // trigger context
  "output": { ... },       // results per action
  "error": null,
  "startedAt": "...",
  "finishedAt": "..."
}
```

View via: `GET /workflows/:id/executions`.

Testing a Workflow

```
POST /api/v1/workflows/:id/execute-on-record
{
  "recordId": "record-cuid"
}
```

Immediately executes the workflow against the specified record, bypassing the trigger condition. Useful for debugging.

Blueprint Tasks vs. Workflows

Feature	Workflow	Blueprint
Trigger	Event-based	Stage transition
Approval	No	Yes — human task

Feature	Workflow	Blueprint
Field lock	No	Yes — per stage
Visualization	Execution log	Stage tree
Use case	Auto-send email	Multi-stage approval process

Use **Workflows** for automated data manipulation and notifications. Use **Blueprints** when human review/approval is required between stages.

Blueprint (Stage Workflow)

Blueprints define a stage machine for records (e.g., Draft → Review → Approved → Published).

Structure

```
{
  "phases": [
    { "id": "draft", "name": "Draft", "color": "#gray" },
    { "id": "review", "name": "Under Review", "color": "#blue" },
    { "id": "approved", "name": "Approved", "color": "#green" }
  ],
  "transitions": [
    {
      "id": "t1",
      "from": "draft",
      "to": "review",
      "name": "Submit for Review",
      "requiresApproval": true,
      "assignedRole": "MANAGER"
    }
  ],
  "fieldLocks": {
    "review": ["amount", "description"] // locked fields in this stage
  }
}
```

Executing a Transition

```
POST /api/v1/blueprints/execute-transition
{
  "recordId": "...",
  "transitionId": "t1",
  "comment": "Ready for review"
}
```

If `requiresApproval`, a `BlueprintTask` is created for the assigned role. The approver sees it in their task queue (`GET /blueprints/my-pending-tasks`).

Approval/Rejection

```
POST /api/v1/blueprints/pending-tasks/:id/action
{
  "action": "approve",    // or "reject"
  "comment": "Looks good"
}
```